EXOS INPUT_EVALUATOR Function Specification

Version 1.0 · March 2026 Cross-referenced to: Scenario Data Requirements v2 · Field Methodology v1 · Security Posture Report

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1. Purpose & Problem Statement

EXOS currently evaluates input completeness using a **quantity-only check**: if all textarea fields contain content, the system reports the input as valid. This creates a critical quality gap — a user who fills every block with "asdfgh" or copy-pastes their company's Wikipedia page into all three fields receives the same "ready to submit" signal as a user who provides structured, scenario-relevant data.

The `INPUT_EVALUATOR` function replaces this binary presence-check with a **quality-aware, scenario-specific evaluation** that assesses whether the content in each input block is semantically appropriate, structurally sufficient, and likely to produce a reliable AI output for that scenario.

Design Principles

1. **Fail informatively, not punitively.** The evaluator coaches users toward better input — it does not block submission. Outputs are advisory banners and inline guidance, never hard gates (except for Type 1H critical fields — see §5.2).
2. **Value-leakage framing.** Every quality warning references the scenario's Financial Impact of Gap from Data Requirements v2 — not generic validation errors.
3. **Scenario-specific rules.** There is no universal "good input" definition. Each scenario has its own quality criteria derived from its Minimum Required Input, Enhanced Input, Common Failure Mode, and Field Methodology deviation type.
4. **Lightweight execution.** The evaluator runs client-side before the Sentinel Pipeline is invoked. It must not call external APIs or AI models. All checks are rule-based heuristics.
5. **GDPR-safe.** The evaluator never logs, stores, or transmits user input content. It operates on structural properties of the text (token patterns, numeric presence, keyword matching) without persisting any data.

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2. Architecture Overview

2.1 Execution Context

```
User fills input blocks → INPUT_EVALUATOR runs on-change (debounced 800ms)
                        → Returns EvaluationResult per block
                        → UI renders inline feedback + submission banner
```

```
                    → User submits (or improves input first)
                    → Sentinel Pipeline processes request
```

The evaluator is a **pure function** with no side effects:

```typescript
type EvaluationResult = {
  scenarioId: number;
  overallScore: 'READY' | 'IMPROVABLE' | 'INSUFFICIENT';
  blocks: BlockEvaluation[];
  missingEnhancedInputs: string[];       // List of missing Enhanced Input items
  confidenceFlag: 'HIGH' | 'MEDIUM' | 'LOW';
  financialImpactWarning: string | null;  // From Data Requirements v2
  gdprWarnings: string[];                // Active GDPR guardrail violations detected
};

type BlockEvaluation = {
  blockId: string;                       // e.g. 'industryContext', 'block2', 'block3'
  label: string;                         // Scenario-specific label from Field Methodology
  status: 'PASS' | 'WARN' | 'FAIL';
  qualityChecks: QualityCheck[];
  suggestedAction: string | null;        // Coaching text shown to user
};

type QualityCheck = {
  checkId: string;
  passed: boolean;
  severity: 'CRITICAL' | 'WARNING' | 'INFO';
  message: string;                       // User-facing feedback
};
```

2.2 Evaluation Pipeline (per block)

Each block passes through checks in this order:

1. **Structural Check** — Is the content non-trivial? (not gibberish, not a single word, not copy-pasted boilerplate)
2. **Semantic Relevance Check** — Does the content relate to what this block expects?
3. **Data Completeness Check** — Are the specific data points required by Data Requirements v2 present?
4. **Numeric/Structured Data Check** — For scenarios requiring quantitative inputs, are numbers/percentages/currencies present?
5. **GDPR Guardrail Check** — Does the input contain patterns that violate the scenario's privacy guardrail?

6. **Enhanced Input Advisory** — Which Enhanced Input items appear to be missing?

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3. Universal Quality Checks (All Scenarios)

These checks apply to every input block in every scenario before scenario-specific rules run.

3.1 Anti-Gibberish Detection

Check ID: UNIVERSAL_GIBBERISH **Severity:** CRITICAL **Logic:**

- Calculate the ratio of dictionary words to total words. If < 40%, flag as gibberish.
- Dictionary = top 10,000 English words + procurement domain vocabulary (see §8 Glossary).
- Exempt: blocks where the primary content is expected to be numerical (e.g., spend data, KPI metrics).
- Detect repeated character patterns: if any 3+ character sequence repeats more than 5 times, flag.
- Detect keyboard-mash patterns: sequences like "asdf", "qwerty", "zxcv" trigger immediate FAIL.

User Message: "This field appears to contain placeholder or test content. Please provide your actual procurement data to get a reliable analysis."

3.2 Minimum Length Check

Check ID: UNIVERSAL_MIN_LENGTH **Severity:** WARNING (for optional blocks), CRITICAL (for required blocks) **Logic:**

- Block 1 (Industry Context): minimum 30 words.
- Block 2 (Core Data): minimum 40 words OR 5 structured data points (numbers, bullet items, line breaks with content).
- Block 3 (Enhanced/Parameters): minimum 20 words if present, but no CRITICAL flag if empty (this is typically enhanced input).
- Override: Type 1H scenarios have higher thresholds — see §5.2.

User Message (CRITICAL): "Your [Block Label] needs more detail. Outputs generated from minimal input are typically flagged LOW CONFIDENCE and may not be suitable for decision-making."

User Message (WARNING): "Adding more detail to [Block Label] will improve the depth of your analysis. [Scenario-specific coaching tip]."

3.3 Copy-Paste Boilerplate Detection

Check ID: UNIVERSAL_BOILERPLATE **Severity:** WARNING **Logic:**

- If content across Block 1, Block 2, and Block 3 shares > 60% token overlap (Jaccard similarity on word-level trigrams), flag as likely copy-paste of the same content into multiple blocks.
- If content matches common boilerplate patterns: "Lorem ipsum", "To whom it may concern", "Dear Sir/Madam", "test test test" — flag immediately.

User Message: "It looks like the same content may have been entered in multiple fields. Each block serves a different purpose — [Block 2 Label] should contain [expected content type], while [Block 3 Label] should contain [expected content type]."

3.4 Language Detection

Check ID: `UNIVERSAL_LANGUAGE` **Severity:** INFO **Logic:**

- Detect if the primary language is not English using character frequency analysis (non-Latin scripts, or common non-English stopwords > 30% of total).
- This is informational only — EXOS supports multilingual input but the AI inference is optimised for English.

User Message: "Your input appears to be in [detected language]. EXOS processes all inputs in English for optimal analysis quality. Consider translating key terms."

3.5 PII Pattern Detection (GDPR Pre-Screen)

Check ID: `UNIVERSAL_PII` **Severity:** WARNING **Logic:**

- Regex scan for common PII patterns:
  - Email addresses: `\b[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Z|a-z]{2,}\b`
  - Phone numbers: `\+?\d[\d\s\-\(\)]{7,}\d`
  - National ID patterns (EU formats): sequences matching known VAT ID, social security, or national ID formats
  - Full names preceded by titles: `(Mr|Mrs|Ms|Dr|Prof)\.\s+[A-Z][a-z]+\s+[A-Z][a-z]+`
- This is a pre-screen only. The Sentinel Pipeline's anonymisation layer is the authoritative control (C-2). The evaluator's role is to warn users before submission so they can self-correct.

User Message: "We detected what may be personal information (e.g., email addresses or names). EXOS recommends using role-based references (e.g., 'Operations Lead') instead of individual names. [Scenario-specific GDPR guardrail text]."

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4. Scenario Group Quality Rules

Each scenario group has shared characteristics that define a baseline quality profile. Scenario-specific overrides (§5) layer on top of these group rules.

4.1 Group A — Analytical Value (Scenarios 1–8)

Group Characteristic: Deterministic calculation engine. Missing numerical inputs produce mathematically unreliable outputs.

Group-Level Checks:

Check ID	Logic	Severity
GROUPA_NUMERIC_PRESENCE	Block 2 and/or Block 3 must contain at least 2 parseable numbers (integers, decimals, percentages, or currency values with €/$£ symbols). If zero numbers are found, the AI cannot perform calculations.	CRITICAL
GROUPA_CURRENCY_CONSISTENCY	If multiple currency symbols are detected, warn about potential FX mismatch.	WARNING
GROUPA_TIMEFRAME_PRESENT	At least one temporal reference must exist (years, months, "annual", "per quarter", date ranges). Financial models without a time dimension are incomplete.	WARNING
GROUPA_UNIT_PRESENT	Detect presence of measurement units (%, €, $, units, hours, kg, m², etc.). Analytical scenarios require quantified inputs.	WARNING

Group-Level User Message (CRITICAL): "This scenario performs financial calculations that require numerical data. Your input currently contains no parseable numbers, percentages, or currency values. Without these, the output will be indicative only and should not be used for financial decisions."

4.2 Group B — Workflow & Convenience (Scenarios 9–15)

Group Characteristic: Document generation quality is proportional to contextual completeness. Narrative context is the primary input signal.

Group-Level Checks:

Check ID	Logic	Severity
GROUPB_CONTEXT_DEPTH	Block 1 must contain at least 2 of: industry keyword, organisation size indicator, geographic reference, or regulatory mention. Generic context ("a company needs something") degrades output from structured to generic.	WARNING
GROUPB_SPECIFICITY	Block 2 must contain at least 1 proper noun or specific entity reference (anonymised is fine — "Supplier A" counts, but "some supplier" does not).	WARNING
GROUPB_ACTIONABLE_DETAIL	Block 2 must contain at least one of: a deadline/date, a volume/quantity, a named deliverable, or a concrete requirement.	WARNING

Group-Level User Message (WARNING): "The more specific your input, the more usable the output. Generic descriptions produce generic documents that require heavy manual editing. Consider adding specific timelines, volumes, or compliance standards."

4.3 Group C — Reliability & Compliance (Scenarios 16–20)

Group Characteristic: Risk and legal defence. Incomplete data directly reduces the AI's ability to identify gaps and exposures.

Group-Level Checks:

Check ID	Logic	Severity
GROUPC_REGULATORY_REFERENCE	At least one regulatory standard or legal framework must be mentioned (GDPR, SOC2, ISO, TUPE, IR35, IFRS, etc.) or the compliance analysis will be jurisdiction-agnostic and potentially misleading.	WARNING
GROUPC_RISK_SPECIFICITY	Input must contain at least one specific risk, hazard, or compliance concern — not just a category name.	WARNING
GROUPC_DOCUMENT_COMPLETENESS	For Type 2 scenarios (SOW Critic, Software Licensing Audit): if text input is < 200 words, warn that partial documents receive partial reviews.	WARNING

Group-Level User Message (WARNING): "Compliance and risk scenarios are only as thorough as the input provided. The AI audits what it can see — missing context means missing risk identification. [Financial Impact of Gap for this scenario]."

4.4 Group D — Strategic Mentorship (Scenarios 21–27)

Group Characteristic: Academic frameworks (BATNA, Kraljic, Porter's) applied to real inputs. Vague inputs yield generic strategy.

Group-Level Checks:

Check ID	Logic	Severity
GROUPD_STRATEGIC_SPECIFICITY	Block 2 must contain at least one of: a spend figure, a market structure description, a competitive position reference, or a named strategic concern. "We need a strategy" with no specifics triggers this.	WARNING
GROUPD_ALTERNATIVES_PRESENT	For negotiation/decision scenarios: check if the input references at least one alternative option or comparator. Strategies without alternatives cannot calculate BATNA/ZOPA.	WARNING
GROUPD_TEMPORAL_HORIZON	Check for a planning horizon reference (1-year, 3-year, short-term, long-term). Strategy without a time horizon is tactical advice, not strategic planning.	INFO

Group-Level User Message (WARNING): "Strategic scenarios require specifics to move beyond generic frameworks. Without concrete data points (spend, market structure, alternatives), the output will resemble a textbook rather than an actionable strategy."

4.5 Group E — Real-Time Knowledge (Scenarios 28–29)

Group Characteristic: Powered by Perplexity Sonar Pro live search. Query specificity is the lever for actionable intelligence.

Group-Level Checks:

Check ID	Logic	Severity
GROUPE_QUERY_SPECIFICITY	Block 2 must contain at least one of: a specific industry niche (not just "manufacturing"), a geographic region, or a named entity. Broad queries return surface-level results.	WARNING
GROUPE_NICHE_DEPTH	If Block 2 contains only one generic industry keyword (e.g., "logistics", "IT", "construction") without qualifiers, flag as too broad.	WARNING
GROUPE_ENTITY_FORMAT	For S29 (Pre-Flight Supplier Audit) specifically: check that Block 2 contains what appears to be a formal entity name (capitalised multi-word phrase, possibly with legal suffix: GmbH, Ltd, S.A., B.V., AG, SRL, etc.).	CRITICAL (S29 only)

Group-Level User Message (WARNING): "Real-time intelligence quality depends entirely on query precision. 'Global logistics market' returns publicly available summaries. 'Automotive-grade MLCC capacitor supply in Southeast Asia' returns competitive intelligence."

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5. Scenario-Specific Quality Rules

These rules override or supplement the group-level checks for individual scenarios where the Data Requirements v2 or Field Methodology v1 documents specify unique quality criteria.

5.1 Type 1 Scenarios — Sub-Prompt Validation

For all Type 1 (Sub-Prompts Required) scenarios, the evaluator checks whether the sub-prompt structure has been followed. If sub-prompts are rendered as guided placeholder text within textareas, the evaluator checks if each sub-prompt section has been addressed.

Check ID: TYPE1_SUBPROMPT_COVERAGE **Logic:**

- Parse the block content against the expected sub-prompt sections for this scenario (defined in the scenario config — see §7).
- If the block contains a single unstructured narrative paragraph with no line breaks, bullet points, or labeled sections, and the scenario expects 3+ discrete data items, flag as likely unstructured.
- Count the number of distinct data items provided vs. expected. If < 50% of expected items are identifiable, flag.

Severity: WARNING **User Message:** "This scenario works best when you provide each data point separately. Try using the guided structure: [list the sub-prompt labels for this block]."

5.2 Type 1H Scenarios — Critical Sub-Prompt Enforcement

The 4 Type 1H scenarios are special: omitting sub-prompts doesn't just reduce quality — it produces a **qualitatively different (incorrect) output type**. These require elevated enforcement.

S3: CAPEX vs. OPEX Analysis

Check	Logic	Severity
S3_WACC_PRESENT	Block 3 must contain a percentage value explicitly associated with WACC, discount rate, hurdle rate, or cost of capital. Regex: keyword proximity check for (WACC|discount rate|hurdle rate|cost of capital) within 20 tokens of a % value.	CRITICAL
S3_TAX_RATE_PRESENT	Block 3 must contain a percentage value associated with tax rate, corporate tax, or tax jurisdiction.	CRITICAL
S3_LIFECYCLE_PRESENT	Block 2 must contain a year count or duration for the asset lifecycle.	CRITICAL

CRITICAL User Message: "Without WACC and tax rate data, the NPV and Payback Period calculations cannot be completed. The CFO will reject an incomplete financial model. Please provide: your WACC or internal hurdle rate (%) and corporate tax rate (%)."

Financial Impact Warning: "IFRS 16 lease capitalisation can add 10–20% to reported liabilities. A flawed model risks audit findings."

S14: Supplier Performance Review

Check	Logic	Severity
S14_KPI_NUMERIC	Block 2 must contain at least 2 parseable percentage values (for OTD %, quality reject rate, invoice accuracy).	CRITICAL
S14_SATISFACTION_SCORE	Block 2 must contain a score on a recognisable scale (1–5, 1–10, or percentage).	WARNING
S14_QUALITATIVE_PRESENT	Block 3 must contain at least 1 narrative observation (minimum 15 words of non-numeric content).	WARNING

CRITICAL User Message: "Without specific KPI percentages, the AI cannot generate a scorecard or benchmark against industry standards. It will produce a narrative assessment instead — which is a different output type entirely. Please provide: OTD %, quality reject rate, and invoice accuracy as numbers."

Financial Impact Warning: "Organisations with structured supplier performance programmes achieve 23% better OTD and 15% quality improvement (CIPS, 2023)."

S23: Make vs. Buy Analysis

Check	Logic	Severity
S23_BILATERAL_COST	The input across Block 2 and Block 3 must contain at least 2 distinct cost references — one that can be associated with internal/in-house cost, and one with external/vendor cost. If both cost figures appear in the same block with no structural separation, warn.	CRITICAL
S23_OVERHEAD_INCLUSION	Check for keywords indicating overhead consideration: "overhead", "fully-loaded", "indirect cost", "management cost", "transition", "integration". If none present, warn that direct cost comparison without overhead is the #1 failure mode.	WARNING

CRITICAL User Message: "Make vs. Buy analysis requires separate cost figures for internal (fully-loaded) and external (total including transition) options. Without both sides, the AI cannot perform a bilateral comparison. Deloitte data shows 44% of outsourcing decisions based on incomplete cost analysis are reversed."

S29: Pre-Flight Supplier Audit

Check	Logic	Severity
S29_LEGAL_ENTITY_FORMAT	Block 2 must contain a capitalised multi-word string that resembles a registered legal entity name. Check for legal suffixes: GmbH, Ltd, Plc, S.A., B.V., AG, SRL, A.S., OÜ, Sp. z o.o., etc. If the input looks like a brand name only (single word, no suffix), flag.	CRITICAL
S29_JURISDICTION_PRESENT	Block 2 must contain a country name or ISO country code.	CRITICAL
S29_BRAND_VS_ENTITY_WARNING	If the input contains a well-known brand name without a legal suffix, warn that brand name ≠ legal entity. Holding structures mean the contracting entity may differ from the trading name.	WARNING

CRITICAL User Message: "CRITICAL: You've entered what appears to be a brand name, not a registered legal entity name. Intelligence retrieved for the wrong entity means sanctions screening fails. Please provide the exact registered entity name (e.g., 'ACME Logistics GmbH', not just 'ACME')."

Financial Impact Warning: "Onboarding a sanctioned supplier carries up to €5M in regulatory fines (EU Sanctions Regulation)."

5.3 Type 2 Scenarios — Document Upload Validation

For the 3 Type 2 scenarios (S5: Spend Analysis, S16: SOW Critic, S19: Software Licensing Audit), Block 2 is architecturally inadequate as a textarea. Until file upload is implemented, these scenarios need additional textarea validation.

S5: Spend Analysis

Check	Logic	Severity
S5_STRUCTURED_DATA	Block 2 should contain tabular or structured data: check for delimiters (commas, tabs, pipes), consistent line structures, or column-header patterns. If pure narrative prose > 200 words with no structured elements, warn.	WARNING
S5_REQUIRED_COLUMNS	Check for presence of keywords mapping to required columns: "supplier" or "vendor", "amount" or "spend" or "€" or "$", "date" or "period", "description" or "item" or "category". Must find ≥ 3 of 4.	CRITICAL
S5_VOLUME_THRESHOLD	If fewer than 5 distinct line items are identifiable, warn that spend analysis requires meaningful transaction volume for classification accuracy.	WARNING

S16: SOW Critic

Check	Logic	Severity
S16_DOCUMENT_LENGTH	If Block 2 content < 200 words, warn that partial SOW pastes receive partial reviews. The AI audits what exists, not what is absent.	WARNING
S16_ENGAGEMENT_TYPE	Check that Block 1 or Block 2 contains an engagement type reference: "fixed-price", "T&M", "time and materials", "milestone", "retainer", "call-off".	WARNING
S16_COMPLETENESS_DISCLAIMER	If Block 2 < 500 words, always surface: "If this is a partial paste of your SOW, consider uploading the complete document (when available) for a more thorough review."	INFO

S19: Software Licensing Audit

Check	Logic	Severity
S19_METRIC_TYPE_PRESENT	Check for licence metric keywords: "named user", "concurrent", "CPU", "core", "revenue", "device", "seat", "subscription", "per-user". If none found, the most common compliance gap (metric mismatch) cannot be assessed.	CRITICAL
S19_EXPIRY_DATE_PRESENT	Check for a date or temporal reference associated with renewal, expiry, or true-up.	WARNING
S19_USAGE_METRICS	Check for current usage numbers (user count, device count, revenue figure). Without usage vs. entitlement comparison, compliance assessment is impossible.	WARNING

5.4 Remaining Scenario-Specific Overrides

The following scenarios have specific quality requirements beyond their group and type defaults.

S1: TCO Analysis

Check	Severity	Logic
S1_LIFECYCLE_DURATION	CRITICAL	Block 2 must contain a year-count or duration reference. TCO without a lifecycle horizon is a spot-price comparison, not TCO.
S1_OPEX_CATEGORIES	WARNING	Block 2 or Block 3 should mention at least 2 OPEX categories (maintenance, training, logistics, disposal, support, energy, insurance). Single-category TCO misses 30–60% of true cost.
S1_VENDOR_COMPARISON	INFO	If only 1 vendor/option is referenced, advise that comparative TCO (current vs. proposed) yields stronger decision support.

S2: Cost Breakdown / Should-Cost

Check	Severity	Logic
S2_MATERIAL_REFERENCE	WARNING	Block 2 must mention at least 1 material category or component family. Without material input, should-cost decomposition defaults to high-level industry averages.
S2_GEOGRAPHY_PRESENT	WARNING	A manufacturing region or supplier geography is required for labour-rate benchmarking. Check for country names or regional references.

S4: Savings Calculation

Check	Severity	Logic
S4_BASELINE_PRESENT	CRITICAL	Must contain a historical baseline price or spend figure. Savings cannot be calculated without a reference point.
S4_NEW_PRICE_PRESENT	CRITICAL	Must contain a new/negotiated price or target.
S4_SAVINGS_CATEGORY	WARNING	Check for hard/soft/avoidance classification keywords. Mixing categories is the #1 reason Finance rejects savings reports.

S6: Forecasting & Budgeting

Check	Severity	Logic
S6_HISTORICAL_DATA	CRITICAL	Must reference at least 2 years of historical spend or volume data.
S6_PLANNING_HORIZON	WARNING	Must specify 1-year or 3-year planning horizon.
S6_MACRO_FACTORS	WARNING	Should mention at least 1 macro factor (inflation, CPI, FX, energy, commodity). Flat-line forecasts without macro adjustment are the documented failure mode.

S7: SaaS Optimisation

Check	Severity	Logic
S7_TOOL_LIST	CRITICAL	Must list at least 2 SaaS tools with identifiable names. A single tool is an audit, not a portfolio optimisation.
S7_LICENCE_COUNTS	WARNING	Should contain numeric licence counts. Without counts, utilisation analysis is impossible.
S7_RENEWAL_DATES	WARNING	At least 1 renewal date or expiry reference. Missing the 90-day cancellation window is the documented failure mode.

S8: Specification Optimisation

Check	Severity	Logic
S8_SPEC_DETAIL	WARNING	Must contain at least 1 specific technical parameter (tolerance, grade, material, standard reference). Generic "reduce costs" without spec detail yields generic advice.

S9: RFP Generator

Check	Severity	Logic
S9_TIMELINE_PRESENT	WARNING	Must contain a delivery timeline or deadline reference.
S9_COMPLIANCE_STANDARD	WARNING	Should mention at least 1 regulatory/compliance standard. RFPs without compliance clauses are the documented reissue trigger.

S10: SLA Definition

Check	Severity	Logic
S10_UPTIME_REQUIREMENT	WARNING	Must contain an availability/uptime target or critical failure definition.
S10_PENALTY_MECHANISM	INFO	Check for penalty, credit, or remedy keywords. SLAs without financial penalties are the documented failure mode.

S11: Tail Spend Sourcing

Check	Severity	Logic
S11_ITEM_DESCRIPTION	WARNING	Must describe a specific item or service, not just a category.
S11_QUANTITY_DATE	WARNING	Should contain a quantity and delivery date/timeline.

S12: Contract Template

Check	Severity	Logic
S12_AGREEMENT_TYPE	CRITICAL	Must specify an agreement type (supply, services, NDA, framework, lease).
S12_JURISDICTION	CRITICAL	Must specify a governing law jurisdiction. Jurisdiction mismatch is the documented failure mode.
S12_COMMERCIAL_TERMS	WARNING	Should mention at least 1 commercial term (payment terms, liability cap, warranty period).

S13: Requirements Gathering (BRD)

Check	Severity	Logic
S13_REQUIREMENT_LIST	CRITICAL	Block 2 must contain what appears to be a list (multiple line breaks, bullet-like patterns, numbered items, or "and"/"or" conjunctions linking distinct items). A single paragraph with no list structure is flagged.
S13_GOAL_STATEMENT	WARNING	Should contain a project goal or business problem statement.

S15: Procurement Project Planning

Check	Severity	Logic
S15_MILESTONES	WARNING	Must contain at least 2 milestone or phase references with associated durations.
S15_STAKEHOLDER_ROLES	WARNING	Should mention at least 1 stakeholder role. The missing-RACI failure mode is the most common cause of project delays.

S17: Risk Assessment

Check	Severity	Logic
S17_RISK_COUNT	WARNING	Input should describe at least 2 distinct risks or hazards. A single-risk assessment doesn't constitute a risk register.
S17_REGULATORY_CONTEXT	WARNING	Should mention the regulatory environment (GDPR, financial services, manufacturing).

S18: Risk Matrix

Check	Severity	Logic
S18_RISK_LIST_FORMAT	CRITICAL	Must contain at least 5 identifiable risk items. The matrix cannot be generated with fewer (per Field Methodology).
S18_PROBABILITY_IMPACT	WARNING	Should contain probability and impact ratings (H/M/L or numeric). Without these, the AI defaults to generic ratings with low credibility.

S20: Category Risk Evaluation

Check	Severity	Logic
S20_CATEGORY_SPECIFICS	WARNING	Must name a specific category (not just "procurement").
S20_SUPPLIER_COUNT	WARNING	Should contain a supplier count or concentration metric.
S20_GEOGRAPHY	WARNING	Must reference a supply chain geography.

S21: Negotiation Preparation

Check	Severity	Logic
S21_SUPPLIER_PROPOSAL	CRITICAL	Must reference a supplier's initial position (price, terms, or offer).
S21_TARGET_OUTCOME	CRITICAL	Must state the buyer's target outcome.
S21_ALTERNATIVE	WARNING	Should reference at least 1 alternative. Without alternatives, BATNA calculation is impossible.

S22: Category Strategy

Check	Severity	Logic
S22_ANNUAL_SPEND	WARNING	Should contain an annual spend figure for the category.
S22_MARKET_STRUCTURE	WARNING	Should describe the supply market (competitive, oligopoly, specialist).

S24: Volume Consolidation

Check	Severity	Logic
S24_MULTI_SUPPLIER	CRITICAL	Must reference at least 3 suppliers or spend sources. Consolidation with fewer than 3 is not a consolidation exercise.
S24_SPEND_SPLIT	WARNING	Should contain spend amounts or percentages per supplier.

S25: Supplier Dependency Planner

Check	Severity	Logic
S25_SYSTEM_LIST	WARNING	Must list at least 1 system, contract, or vendor with integration/switching risk.
S25_SWITCHING_COST	INFO	If no switching cost estimate is provided, advise that technical switching costs are routinely underestimated by 300–500%.

S26: Disruption Management

Check	Severity	Logic
S26_DISRUPTION_TYPE	CRITICAL	Must specify the nature of the disruption (supplier failure, logistics, geopolitical, force majeure).
S26_AFFECTED_LINES	WARNING	Should specify affected product lines or categories.
S26_INVENTORY_BUFFER	WARNING	Should mention current inventory or stock buffer. Without this, the response plan cannot be calibrated to urgency.

S27: Black Swan Scenario Simulation

Check	Severity	Logic
S27_SCENARIO_TYPE	CRITICAL	Must specify a scenario type (pandemic, natural disaster, geopolitical embargo, cyberattack).
S27_SUPPLY_NODES	WARNING	Should describe at least 2 supply chain nodes (suppliers, routes, production sites).
S27_BCP_STATUS	INFO	Check for BCP/business continuity references. Absence triggers an advisory.

S28: Market Snapshot

Check	Severity	Logic
S28_NICHE_SPECIFICITY	WARNING	Must contain a specific industry niche, not just a top-level sector. "Manufacturing" is too broad; "automotive-grade MLCC capacitors" is specific. Heuristic: if the industry reference is a single common word, flag.
S28_GEOGRAPHIC_FOCUS	WARNING	Should specify a target geographic region.

6. Scoring Logic & UI Behaviour

6.1 Overall Score Calculation

The overallScore is derived from the most severe failing check:

Condition	Overall Score
Any CRITICAL check has passed: false	INSUFFICIENT
Any WARNING check has passed: false (but all CRITICAL pass)	IMPROVABLE
All CRITICAL and WARNING checks pass	READY

6.2 Confidence Flag Derivation

The confidenceFlag combines the AI Confidence Dependency level (from Data Requirements v2) with the evaluation result:

Scenario Confidence Dependency	Overall Score = READY	Overall Score = IMPROVABLE	Overall Score = INSUFFICIENT
HIGH	HIGH	MEDIUM	LOW
MEDIUM-HIGH	HIGH	MEDIUM	LOW
MEDIUM	MEDIUM	LOW	LOW

When `confidenceFlag` is `LOW`, the output watermark `LOW CONFIDENCE — KEY DATA MISSING` must be applied to the AI output. This watermark must persist in all export formats (Excel, PDF, Jira) per Data Requirements v2 Appendix.

6.3 UI Rendering Rules

Overall Score	Submission Banner	Banner Colour	Can Submit?
`READY`	"Your input is well-structured for this scenario."	Green	Yes
`IMPROVABLE`	"Your analysis will be more reliable with additional detail. [First WARNING message]. Estimated value at risk: [Financial Impact of Gap]."	Amber/Orange	Yes (with advisory)
`INSUFFICIENT`	"Critical data is missing for a reliable analysis. [First CRITICAL message]. Without this data: [Financial Impact of Gap]."	Red	Yes (with prominent warning)

Important: The evaluator never hard-blocks submission. Even `INSUFFICIENT` allows the user to proceed — but the AI output carries the `LOW CONFIDENCE` watermark. The only exception would be if the input is pure gibberish (< 40% real words), in which case the submit button may be visually de-emphasised but never disabled.

6.4 Inline Feedback Positioning

- Each block displays its `status` indicator (green check / amber warning / red alert icon) adjacent to the block label.
- The `suggestedAction` text appears below the textarea as helper text, only when `status` is WARN or FAIL.
- GDPR warnings appear as a distinct callout with a shield icon, separate from quality feedback.
- Enhanced Input advisories appear as a collapsible "Improve your results" section below the submit button, listing the missing Enhanced Input items.

---

7. Scenario Configuration Schema

Each scenario must have a configuration object that the evaluator consumes. This configuration is the **single source of truth** for all scenario-specific checks.

```typescript
```

```typescript
type ScenarioEvalConfig = {
  scenarioId: number;
  scenarioName: string;
  group: 'A' | 'B' | 'C' | 'D' | 'E';
  deviationType: 0 | 1 | '1H' | 2;
  confidenceDependency: 'HIGH' | 'MEDIUM-HIGH' | 'MEDIUM';

  blocks: {
    blockId: string;
    label: string;                    // From Field Methodology v1
    isRequired: boolean;
    minWords: number;                 // Override for UNIVERSAL_MIN_LENGTH
    expectedDataType: 'narrative' | 'numeric' | 'structured' | 'document';
    subPrompts?: string[];            // For Type 1/1H: the guided placeholder labels
  }[];

  scenarioChecks: {
    checkId: string;
    targetBlock: string;              // blockId this check applies to
    severity: 'CRITICAL' | 'WARNING' | 'INFO';
    checkType: 'keyword_presence' | 'numeric_presence' | 'pattern_match'
          | 'list_structure' | 'entity_format' | 'min_items';
    params: Record<string, any>;      // Check-type-specific parameters
    failMessage: string;              // User-facing message
    coachingTip?: string;             // Optional additional guidance
  }[];

  financialImpactGap: string;         // Exact text from Data Requirements v2
  commonFailureMode: string;          // Exact text from Data Requirements v2
  gdprGuardrail: string;              // Exact text from Data Requirements v2

  enhancedInputItems: string[];       // List of Enhanced Input items to check for
};
```

7.1 Example Configuration: S3 — CAPEX vs. OPEX Analysis

```typescript
```

```
const S3_CONFIG: ScenarioEvalConfig = {
  scenarioId: 3,
  scenarioName: "CAPEX vs. OPEX Analysis",
  group: 'A',
  deviationType: '1H',
  confidenceDependency: 'HIGH',

  blocks: [
    {
      blockId: 'industryContext',
      label: 'Industry & Business Context',
      isRequired: true,
      minWords: 30,
      expectedDataType: 'narrative',
    },
    {
      blockId: 'block2',
      label: 'Asset or Acquisition Details',
      isRequired: true,
      minWords: 40,
      expectedDataType: 'numeric',
      subPrompts: [
        'Asset description',
        'Financial lifespan (years)',
        'Annual lease / subscription price (€)',
        'Estimated purchase price (€)',
        'Preferred depreciation method (straight-line or declining)',
      ],
    },
    {
      blockId: 'block3',
      label: 'Financial Parameters',
      isRequired: true,       // 1H = sub-prompts are critical
      minWords: 20,
      expectedDataType: 'numeric',
      subPrompts: [
        'WACC or internal hurdle rate (%)',
        'Corporate tax rate (%)',
        'Maintenance and insurance costs (€/year)',
        'Residual / salvage value estimate (€)',
        'Off-balance-sheet lease impact notes (IFRS 16)',
      ],
    },
```

```
],

scenarioChecks: [
  {
    checkId: 'S3_WACC_PRESENT',
    targetBlock: 'block3',
    severity: 'CRITICAL',
    checkType: 'keyword_presence',
    params: {
      keywords: ['WACC', 'discount rate', 'hurdle rate', 'cost of capital'],
      requireNumericProximity: true,
      proximityTokens: 20,
    },
    failMessage: 'WACC or discount rate is missing. Without this, NPV and Payback Period cannot be calculated.',
    coachingTip: 'Your WACC is typically available from your Finance team. If unknown, a common EU mid-market range is
  },
  {
    checkId: 'S3_TAX_RATE_PRESENT',
    targetBlock: 'block3',
    severity: 'CRITICAL',
    checkType: 'keyword_presence',
    params: {
      keywords: ['tax rate', 'corporate tax', 'Körperschaftsteuer', 'impôt sur les sociétés'],
      requireNumericProximity: true,
      proximityTokens: 20,
    },
    failMessage: 'Corporate tax rate is missing. Tax treatment is a critical differentiator between CAPEX and OPEX structures
  },
  {
    checkId: 'S3_LIFECYCLE_PRESENT',
    targetBlock: 'block2',
    severity: 'CRITICAL',
    checkType: 'keyword_presence',
    params: {
      keywords: ['years', 'year', 'lifespan', 'lifecycle', 'duration', 'term'],
      requireNumericProximity: true,
      proximityTokens: 10,
    },
    failMessage: 'Asset lifecycle duration is missing. The financial model requires a time horizon to calculate NPV.',
  },
],

financialImpactGap: "IFRS 16 lease capitalisation can add 10–20% to reported liabilities. A flawed model risks audit finding
commonFailureMode: "Without WACC and tax data, NPV and Payback Period calculations are mathematically incomplete.
```

```
  gdprGuardrail: "Do not include banking covenants, credit facility terms, or exact corporate tax identifiers. Use 'Tax Rate: ~2

  enhancedInputItems: [
    'WACC / internal hurdle rate',
    'Corporate tax rate (jurisdiction)',
    'Maintenance and insurance costs',
    'Residual / salvage value estimate',
    'Off-balance-sheet lease impact (IFRS 16)',
  ],
};
```

---

8. Procurement Domain Glossary

The following terms supplement the standard English dictionary for the gibberish detection check. These are domain-specific terms that should be recognised as valid vocabulary:

Financial: TCO, OPEX, CAPEX, NPV, WACC, IRR, ROI, P&L, EBITDA, FX, CPI, PPI, IFRS, depreciation, amortisation, hurdle rate, payback, escalation, inflation, NPV, salvage, residual, true-up

Procurement: RFP, RFQ, RFI, BRD, SOW, SLA, KPI, OTD, NDA, TUPE, IR35, BATNA, ZOPA, Kraljic, maverick spend, tail spend, framework agreement, call-off, T&M, retainer, spot-buy, off-contract

Compliance: GDPR, SOC2, ISO, CSRD, ESG, MAR, sanctions, embargo, due diligence, PII, anonymisation, DPA, DPIA, consent, data minimisation

Supply Chain: BOM, lead time, MOQ, safety stock, dual-source, single-source, tier-1, tier-2, incoterms, landed cost, logistics, freight, customs, tariff

Taxonomy: UNSPSC, eCl@ss, cost-centre, category, sub-category, spend cube

Legal Suffixes: GmbH, Ltd, Plc, S.A., B.V., AG, SRL, A.S., OÜ, "Sp. z o.o.", NV, SE, KG, AB, Oy, ApS

---

9. Enhanced Input Advisory Logic

When the overall score is `READY` or `IMPROVABLE`, the evaluator should scan for missing Enhanced Input items and surface them as a coaching opportunity — not a warning.

Logic per Enhanced Input item:

For each item in `enhancedInputItems`, check if related keywords are present in any block. If absent, add to `missingEnhancedInputs[]`.

UI Rendering:

A collapsible "Unlock deeper analysis" section appears below the submission area:

> **Your input meets the minimum threshold for this scenario. You can improve your results by adding:**
> - ☐ [Missing Enhanced Input 1] — [1-line explanation of why it helps]
> - ☐ [Missing Enhanced Input 2] — [1-line explanation of why it helps]

This section is hidden when `overallScore === 'INSUFFICIENT'` (priority is on fixing critical gaps, not upselling enhanced inputs).

---

10. GDPR Guardrail Integration

Each scenario has a specific GDPR guardrail from Data Requirements v2. The evaluator surfaces these as contextual privacy reminders.

Trigger Logic:

- If `UNIVERSAL_PII` check fires, append the scenario-specific GDPR guardrail text.
- If the scenario is in a group with elevated GDPR sensitivity (Group A financial data, Group C compliance documents), always show a privacy reminder banner in the UI regardless of whether PII was detected.
- For S5 (Spend Analysis), always show the supplier anonymisation reminder since raw spend data upload is the primary input.

Privacy Reminder Banner (contextual, non-blocking):

> 🛡️ **Privacy Reminder:** [Scenario-specific GDPR guardrail text from Data Requirements v2]. EXOS anonymises your data before AI processing, but we recommend removing sensitive details before input.

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11. Implementation Priority

Priority	Component	Rationale
P0	Universal checks (§3) + scoring logic (§6)	Catches the "all fields filled with nonsense" problem immediately. No scenario-specific config needed.
P1	Type 1H critical checks (§5.2) — S3, S14, S23, S29	These 4 scenarios produce incorrect output types without critical fields. Highest user impact.
P2	Group-level checks (§4)	Covers all 29 scenarios with reasonable heuristics before individual tuning.

Priority	Component	Rationale
P3	Scenario-specific overrides (§5.1, §5.3, §5.4)	Fine-tuning for individual scenarios. Roll out incrementally.
P4	Enhanced Input advisory (§9)	Coaching/upsell feature. Not blocking.
P5	Scenario config objects for all 29 scenarios (§7)	Data entry task. Can be parallelised.

---

12. Testing Strategy

12.1 Test Scenarios per Quality Level

For each scenario, create 4 test inputs:

1. **Gibberish Input** — Random characters, keyboard mash, lorem ipsum → Expected: INSUFFICIENT, all CRITICAL checks fail.
2. **Minimal Valid Input** — Bare minimum data that meets Minimum Required Input → Expected: IMPROVABLE (missing Enhanced Inputs).
3. **Full Quality Input** — Complete Minimum + Enhanced Input → Expected: READY.
4. **Copy-Paste Input** — Same paragraph pasted into all 3 blocks → Expected: IMPROVABLE, boilerplate warning fires.

12.2 Type 1H Regression Tests

For S3, S14, S23, and S29, create additional test cases:

- Input with all narrative, no numbers → Expected: CRITICAL check fails.
- Input with numbers but wrong context (e.g., phone numbers instead of KPIs) → Expected: CRITICAL check fails.
- Input that provides the critical field (e.g., WACC) but buries it in unrelated text → Expected: CRITICAL check passes (keyword proximity heuristic works).

12.3 GDPR Edge Cases

- Input containing obvious PII (full names, emails, phone numbers) → Expected: PII warning fires.
- Input using role-based references correctly ("Operations Lead", "Supplier A") → Expected: no PII warning.

- Input containing legal entity names in S29 → Expected: no PII warning (legal entity names are required for this scenario).

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13. Cross-Reference Index

Document	Section Used	Application in INPUT_EVALUATOR
Data Requirements v2 — Minimum Required Input	Per scenario	Defines what CRITICAL checks enforce
Data Requirements v2 — Enhanced Input	Per scenario	Defines what the advisory section recommends
Data Requirements v2 — Common Failure Mode	Per scenario	Shapes the quality checks that catch the documented failure pattern
Data Requirements v2 — Financial Impact of Gap	Per scenario	Provides the value-leakage framing in user messages
Data Requirements v2 — GDPR Guardrail	Per scenario	Provides scenario-specific privacy reminder text
Data Requirements v2 — AI Confidence Dependency	Per scenario	Determines confidence flag calculation
Data Requirements v2 — Appendix (Implementation Notes)	§6.2, §6.3	Defines the LOW CONFIDENCE watermark and DATA GAP banner requirements
Field Methodology v1 — Deviation Taxonomy	§5.1, §5.2, §5.3	Determines which check tier applies (Type 0/1/1H/2)
Field Methodology v1 — Block Labels & Sub-Prompts	§7 config schema	Provides the field labels and sub-prompt structure
Field Methodology v1 — Appendix A (Type 2 Register)	§5.3	Defines document upload validation rules
Field Methodology v1 — Appendix B (Type 1H Register)	§5.2	Defines the 4 critical scenarios with elevated enforcement
Security Posture Report — C-2	§3.5, §10	PII pre-screen supports but does not replace the anonymisation pipeline

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